ACADEMIC SYLLABUS | Applied Intelligent Systems

Module 1: Foundations of Intelligent Systems

Original test curriculum for automated syllabus parsing. Semester: 2026-27 | Credits: 4 | Course code: AIS204

1.1 Intelligent agents and environments

Learning objective: define the central principles of intelligent agents and environments, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

1.2 Search strategies and complexity

Learning objective: define the central principles of search strategies and complexity, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

1.3 Knowledge representations

Learning objective: define the central principles of knowledge representations, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

1.4 Reasoning with uncertainty

Learning objective: define the central principles of reasoning with uncertainty, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

Assessment: Practice questions, short lab exercise, and written reflection for this module.

ACADEMIC SYLLABUS | Applied Intelligent Systems

Module 2: Data Preparation and Analytics

Original test curriculum for automated syllabus parsing. Semester: 2026-27 | Credits: 4 | Course code: AIS204

2.1 Data collection and quality

Learning objective: define the central principles of data collection and quality, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

2.2 Cleaning and transformation

Learning objective: define the central principles of cleaning and transformation, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

2.3 Feature engineering

Learning objective: define the central principles of feature engineering, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

2.4 Exploratory analysis

Learning objective: define the central principles of exploratory analysis, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

Assessment: Practice questions, short lab exercise, and written reflection for this module.

ACADEMIC SYLLABUS | Applied Intelligent Systems

Module 3: Machine Learning Methods

Original test curriculum for automated syllabus parsing. Semester: 2026-27 | Credits: 4 | Course code: AIS204

3.1 Supervised classification

Learning objective: define the central principles of supervised classification, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

3.2 Regression and metrics

Learning objective: define the central principles of regression and metrics, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

3.3 Unsupervised clustering

Learning objective: define the central principles of unsupervised clustering, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

3.4 Model validation and generalization

Learning objective: define the central principles of model validation and generalization, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

Assessment: Practice questions, short lab exercise, and written reflection for this module.

ACADEMIC SYLLABUS | Applied Intelligent Systems

Module 4: Natural Language Systems

Original test curriculum for automated syllabus parsing. Semester: 2026-27 | Credits: 4 | Course code: AIS204

4.1 Tokenization and language models

Learning objective: define the central principles of tokenization and language models, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

4.2 Text classification

Learning objective: define the central principles of text classification, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

4.3 Information retrieval

Learning objective: define the central principles of information retrieval, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

4.4 Conversational evaluation

Learning objective: define the central principles of conversational evaluation, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

Assessment: Practice questions, short lab exercise, and written reflection for this module.

ACADEMIC SYLLABUS | Applied Intelligent Systems

Module 5: Responsible Deployment

Original test curriculum for automated syllabus parsing. Semester: 2026-27 | Credits: 4 | Course code: AIS204

5.1 Fairness and explainability

Learning objective: define the central principles of fairness and explainability, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

5.2 Model monitoring and drift

Learning objective: define the central principles of model monitoring and drift, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

5.3 Privacy-aware design

Learning objective: define the central principles of privacy-aware design, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

5.4 Deployment and incident response

Learning objective: define the central principles of deployment and incident response, evaluate a worked example and explain a practical application. Class activities include reading, collaborative exercises and a short formative assessment.

Assessment: Practice questions, short lab exercise, and written reflection for this module.